

HE $\rightarrow$ heats outlet at condenser I (molten S) from 150°C to 250°C to inlet at catalytic converter II

* H_2SO_4 Manufacturing

* Properties

MW = 98.08 g/mol

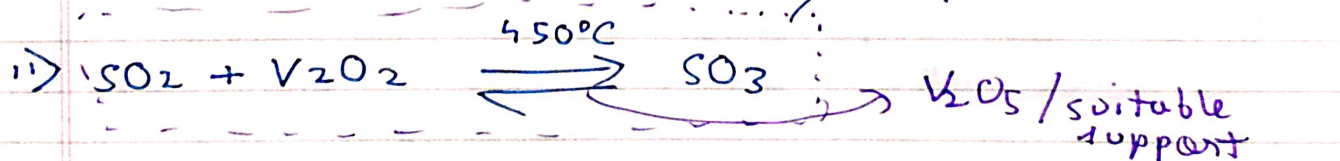
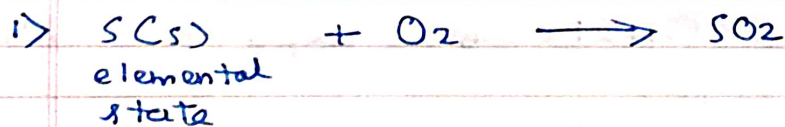
Boiling point = 340°C

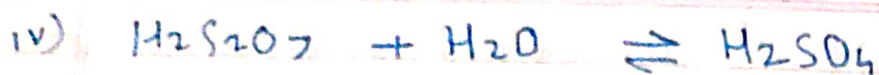
Melting point = -10.5°C

* Solubility $\Rightarrow$ soluble in water $\rightarrow$ produces large amount of heat
 $\Rightarrow \text{SO}_3 \xrightarrow{\text{soluble}} \text{H}_2\text{SO}_4$

* Production of H_2SO_4

① CONTACT PROCESS





→ Catalysts for oxidation of SO_2 to SO_3 .

<u>Current</u>	<u>Previously used</u>
7-8% V_2O_5 loading	
a) $\text{V}_2\text{O}_5 / \text{SiO}_2$	Pt / TiO_2
b) $\text{V}_2\text{O}_5 / \text{SiO}_2 - \text{Al}_2\text{O}_3$	$\text{Pt} / \text{SiO}_2 - \text{Al}_2\text{O}_3$
c) $\text{V}_2\text{O}_5 / \text{Al}_2\text{O}_3$	$\text{Pt} / \text{Al}_2\text{O}_3$
d) $\text{V}_2\text{O}_5 / \text{Kieselguhr}$	$\text{Pt} / \text{Kieselguhr}$

< 0.5 wt% loading

- Disadvantages of Pt based catalyst
- Poisoning Effect (happens when feed is impure)
 - Rapid Heat Deactivation
 - Initial investment is high
 - Fragile

V_2O_5 7-8% loading for 99% conversion, while Pt requires < 0.5% loading for the same 99% conversion

→ Optimized value

- Advantages of V_2O_5 based catalyst
- Cost effective (CH_2PtCl_6 is expensive while NH_4VO_3 is cheaper)
 ↳ ammonium vanadate
 - Immune to poisoning
 - Stable at 450°C

However, there is a disadvantage of V_2O_5 . Feed cannot be concentrated.

$[\text{SO}_2] = 7-10\%$

$[\text{O}_2] = 11-14\%$

SO_2, O_2

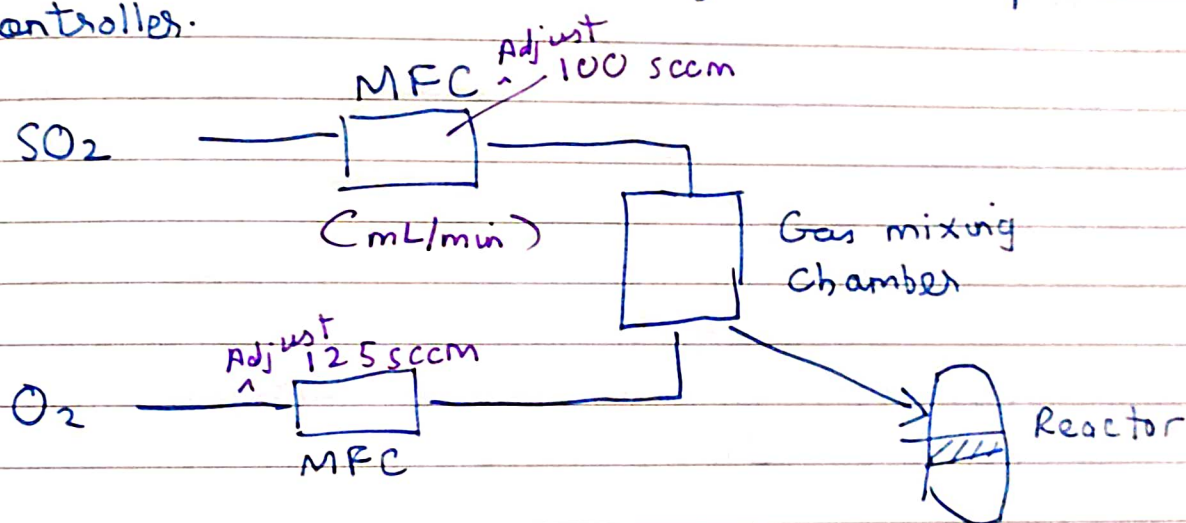


In case of Pt however, it can handle feeds at variable concentration.

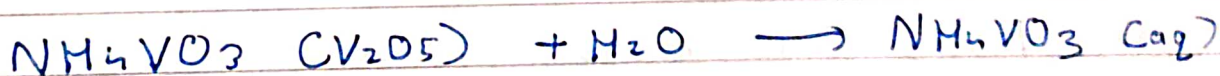
Note $1\% \equiv 10,000 \text{ ppm}$

To achieve $[SO_2] \rightarrow 7 \text{ to } 10\%$ and $[O_2] \rightarrow 11-14\%$, we use SO_2 and O_2 cylinders (99.99% conc).

To reduce the concentration to its optimal value, we use an integral mass flow controllers.



→ V_2O_5 catalyst synthesis on SiO_2 :



Basis: 1g of SiO_2 support $\xrightarrow[\text{mixed}]{\text{Thoroughly}}$ NH_4VO_3/SiO_2 (paste)

glass rod

Muffled furnace
 $550^\circ C$, air
2h

V_2O_5/SiO_2 (powder) $\xleftarrow[\text{Milling}]{\text{Ball}}$ V_2O_5/SiO_2 (clumps)

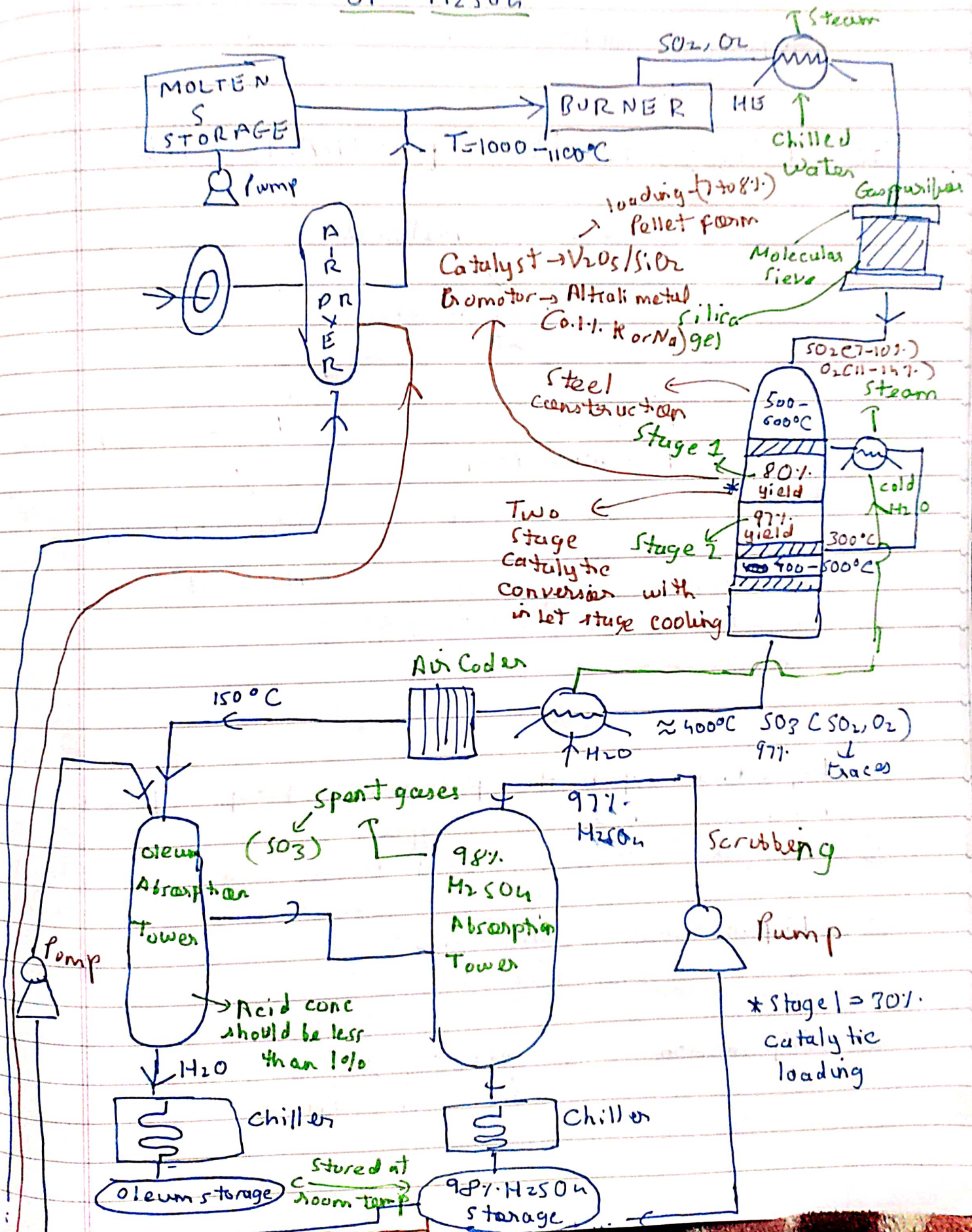
V_2O_5/SiO_2 (pellets) $\xrightarrow[\text{Press}]{\text{Hydrostatic}}$ 3mm diameter

- Same process can be applied for Al_2O_3 as SiO_2 .

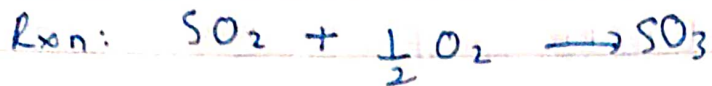
→ Probable Sources of SO_2

- (i) $H_2S \rightarrow SO_2$ CCLAUS PROCESS $\rightarrow$ 1st step
- (ii) Spent $H_2SO_4 / FeSO_4$ (Steel industry)
 $\downarrow \Delta$
 SO_2 (dust particles, acidic impurities)
 $\swarrow$ As, Fe etc.
scrubbing with Monoethanol amine.
 $\searrow$
Cyclone separation,
Electrostatic Precipitation
- (iii) Smelter Sources (non-ferrous materials)
 CuS, PbS, ZnS, MoS_2
 $\downarrow O_2$
 SO_2 (impure) $\rightarrow$ purification needed.
 $\swarrow$ Zinc Blend
 $\searrow$ molybdenum disulphide
- (iv) FeS_2 (40-45% S)
 $\downarrow$ Finnish process
 SO_2 (impure) + hot $\rightarrow 1300^\circ C$ and $1000-1100^\circ C$
 $\downarrow$ cooling $\rightarrow$ Purification $\rightarrow ??$
- (v) Elemental S Oxidation reaction

* PROCESS FLOW DIAGRAM FOR MANUFACTURING OF H_2SO_4

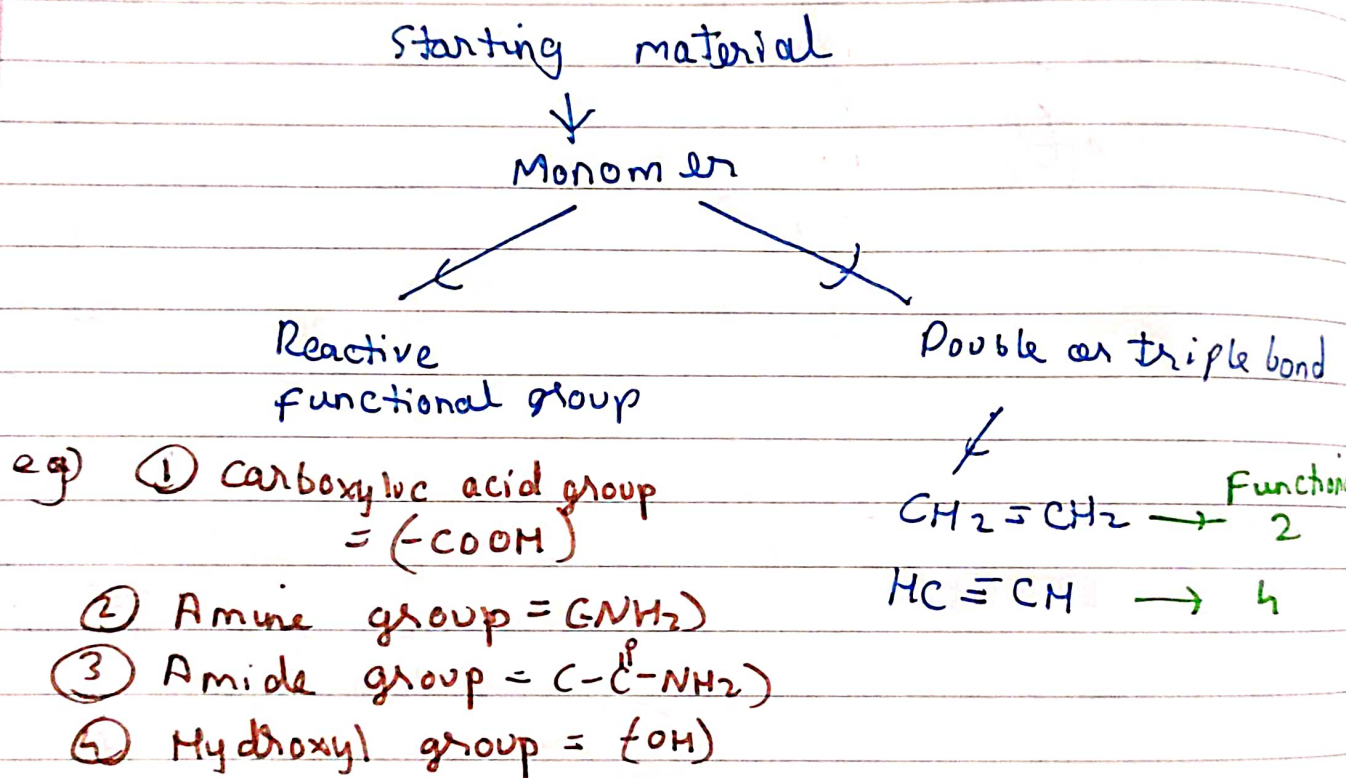


We have a 2 stage reactor as SO_2 when it comes in contact with catalyst causes poisoning of catalyst and its regeneration is time consuming and costly. Also hotspot formation can take place.



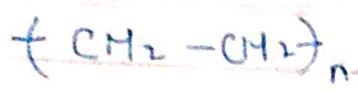
Gas purifier $\rightarrow$ Absorbs moisture (Silica gel)

* POLYMER FUNDAMENTALS



Depending upon the functional groups present, monomer can be of 3 types

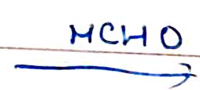
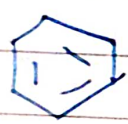
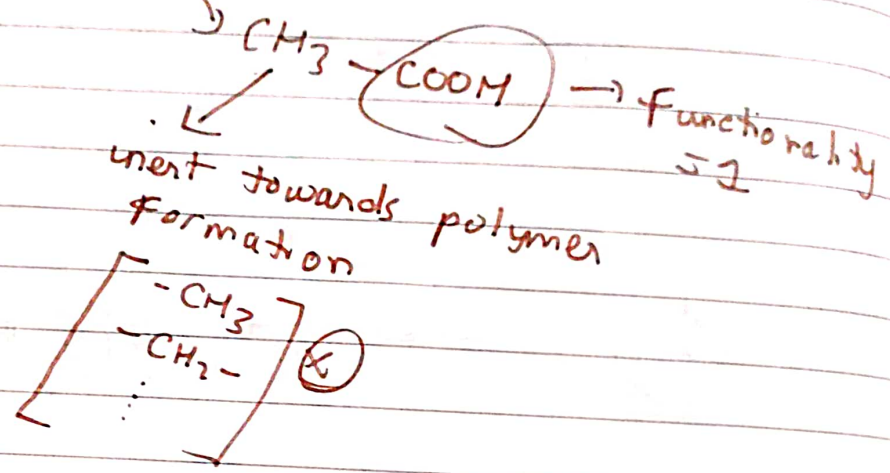
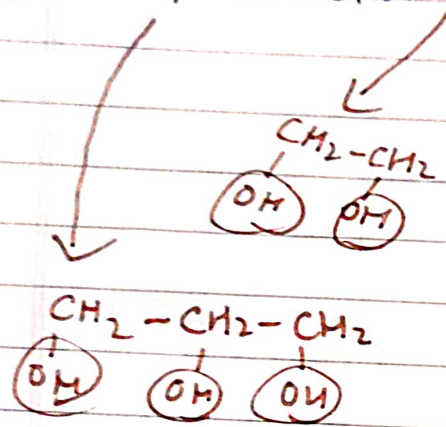
- ① unifunctional $(\text{CH}_3-\text{COOH})$ Acetic acid
inactive towards polymer formation
- ② Bi-functional $(\text{CH}_2-\text{CH}_2)$
Glycol (OH, OH)
- ③ Trifunction $(\text{CH}_2-\text{CH}-\text{CH}_2)$ Glycerol
 $(\text{OH}, \text{OH}, \text{OH})$



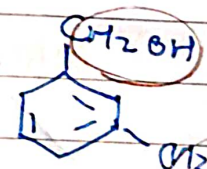
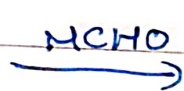
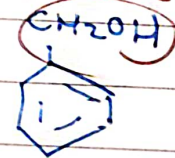
$n \rightarrow$ no. of repeat units

- (i) unifunctional
- (ii) Bifunctional
- (iii) Tri functional

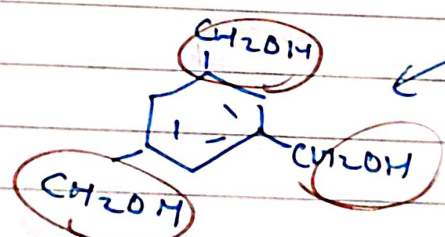
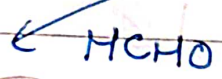
— (X) does not participate in polymer formation.



Monomethylol Benzene



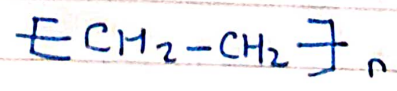
Dimethylol Benzene



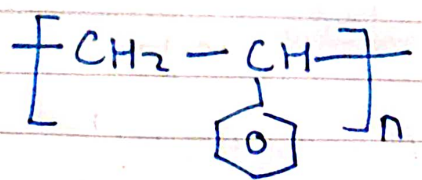
Trimethylol Benzene

* COMMODITY POLYMERS $\rightarrow$ Remember

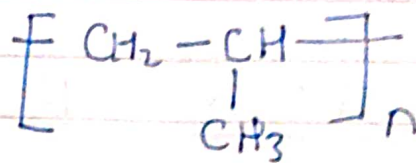
a) Poly ethylene



b) Poly styrene

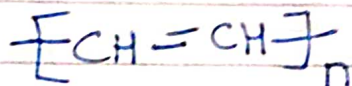


c) Poly Propylene

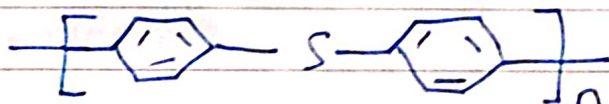


* ELECTRONICS

a) Poly Acetylene

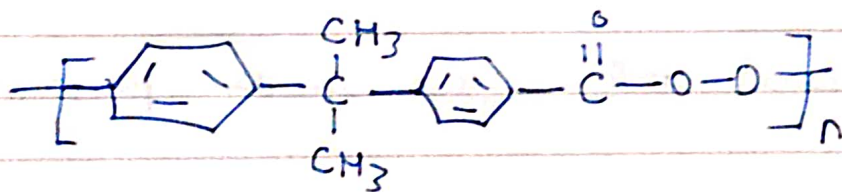


b) Poly Phyneline Sulphide

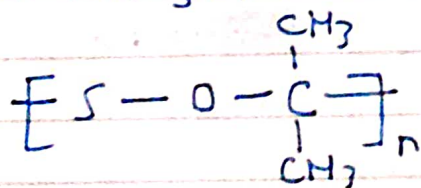


* BIOMEDICAL APPLICATIONS

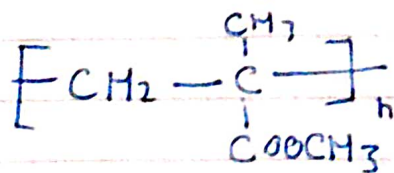
a) Poly Carbonate



b) Silicane Polymers



c) Polymethyl Methacrylate



* CLASSIFICATION OF POLYMERS

① Based on application of heat.

- a) Thermoplasts
- b) Thermosets

a) Thermoplasts

[nylon 6,6
acryl resins
Poly propylene]

Heat → Monomer melts

↓ Cooling
Solidifies

Heating - Cooling - Heating - Cooling - - - - -

↓
Cycle can be continued several times.

No structural deterioration of the polymer takes place.

b) Thermosets

Heating →

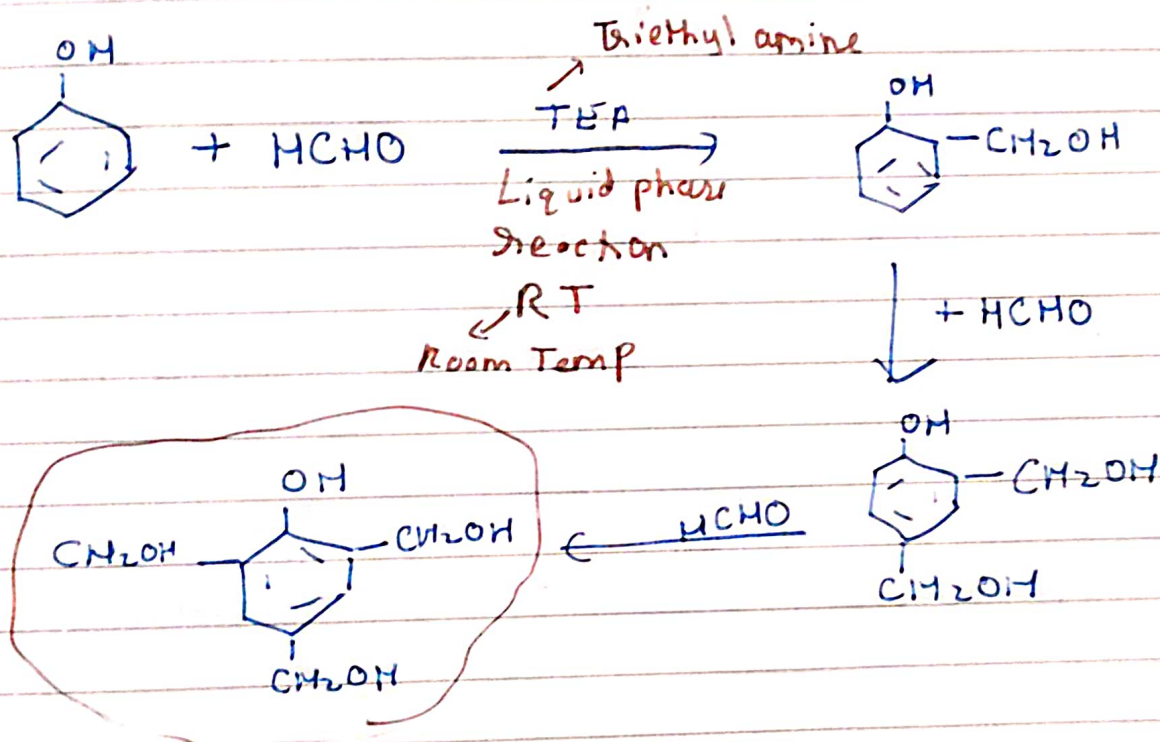
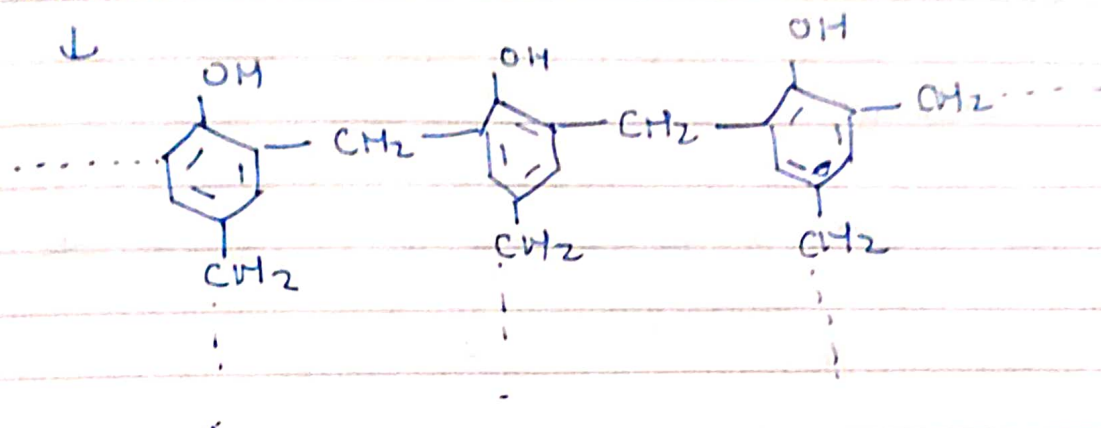
starts ~~deteriorating~~ ^{degrading}

↓ Cooling

Curing of polymer takes place
(Hardening)

eg) Bakelite

Bakelite



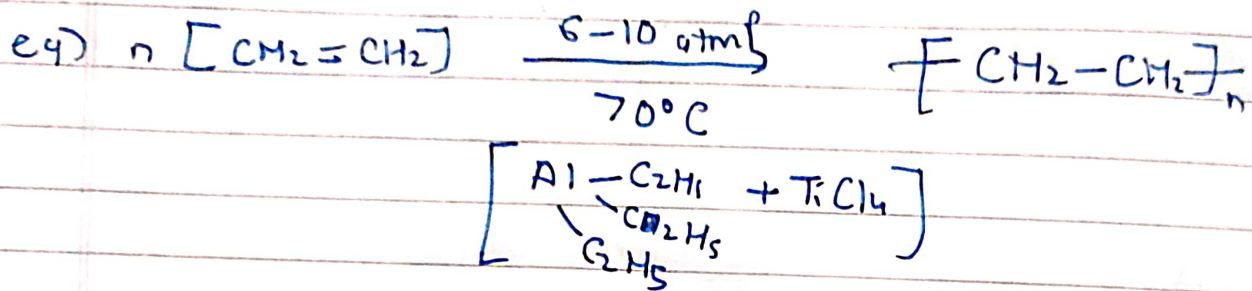
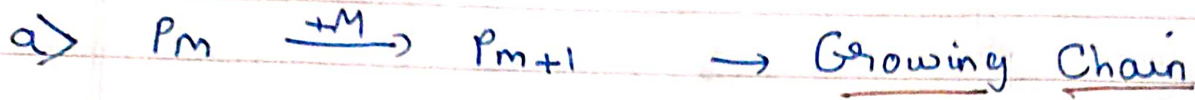
Starting Monomer for Bakelite formation. Leads to formation of Bakelite in acidic medium.
 Alkaline medium → ?

② Based on Reaction Mechanism

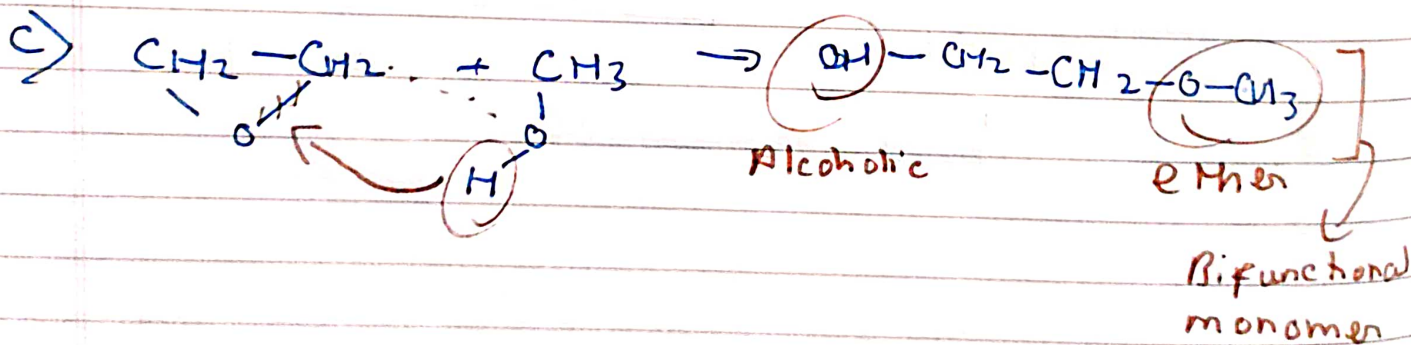
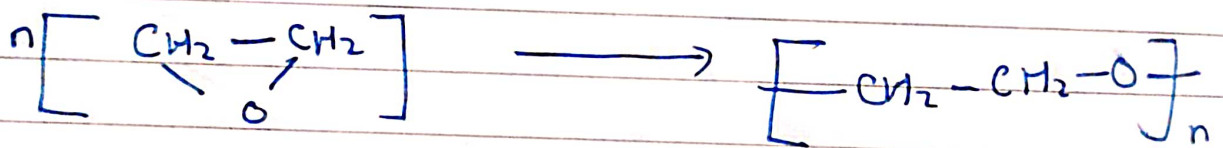
- (i) Addition Polymerization
- ii) Condensation Polymerization

Addition Polymerization

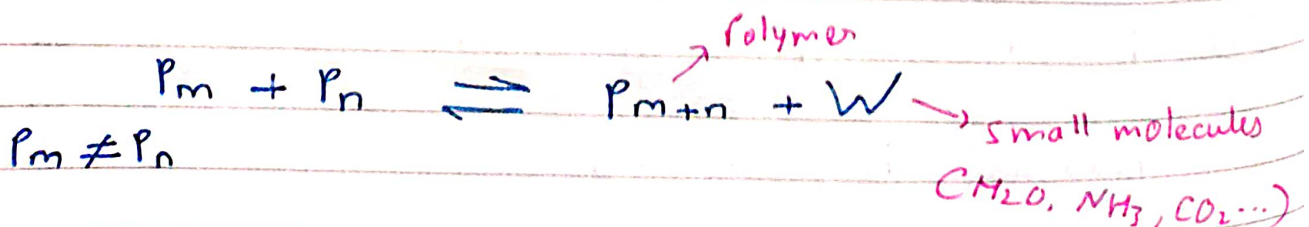
Mechanism

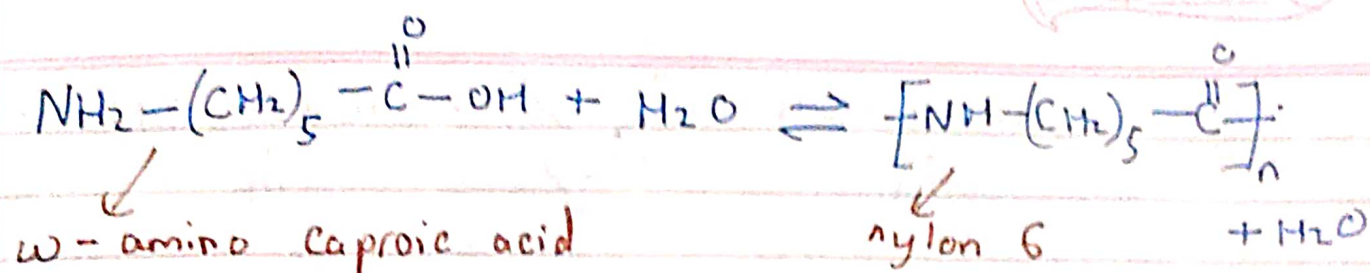


b) Ring Opening Reaction



Condensation Polymerization

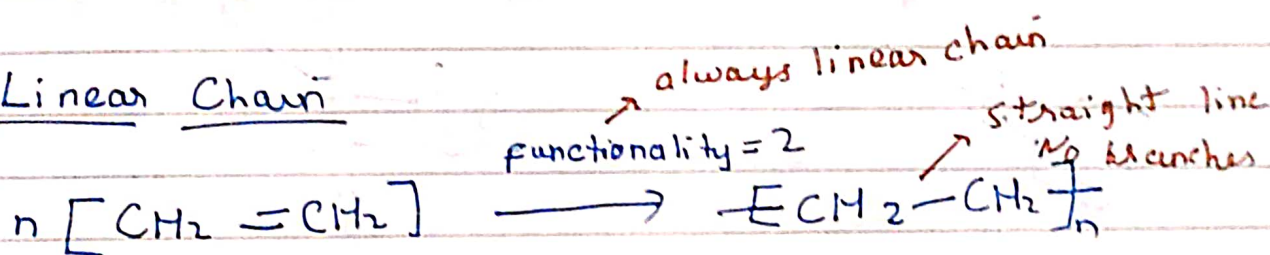




③ Based on Molecular Structure

- (i) Linear Chain Polymer
- ii) Branch Chain Polymer
- iii) Network / Gel Formation

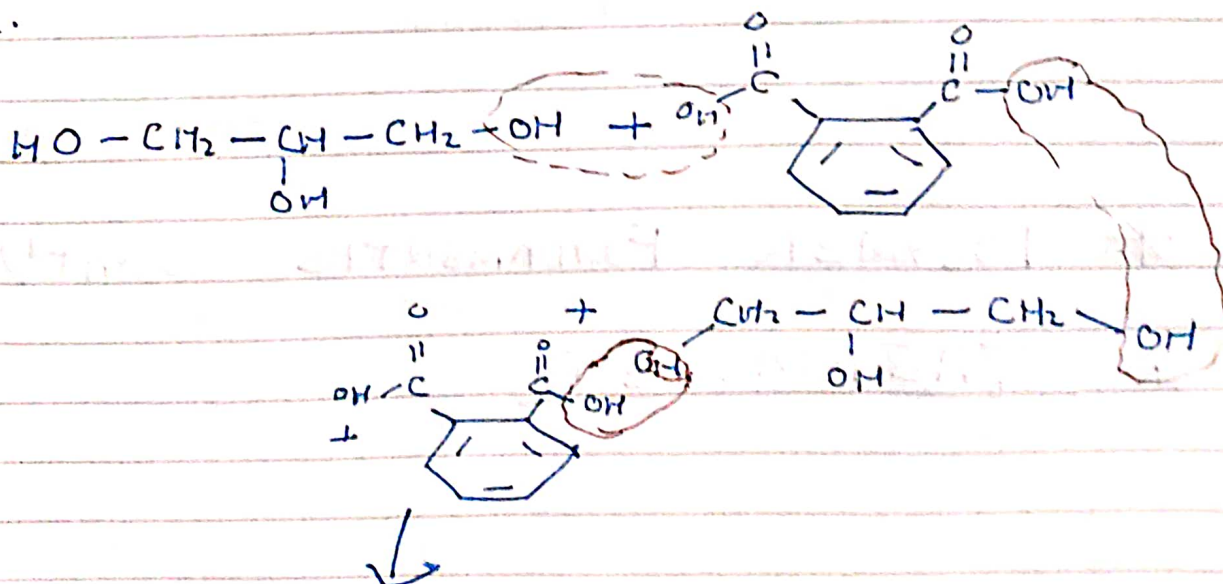
Linear Chain

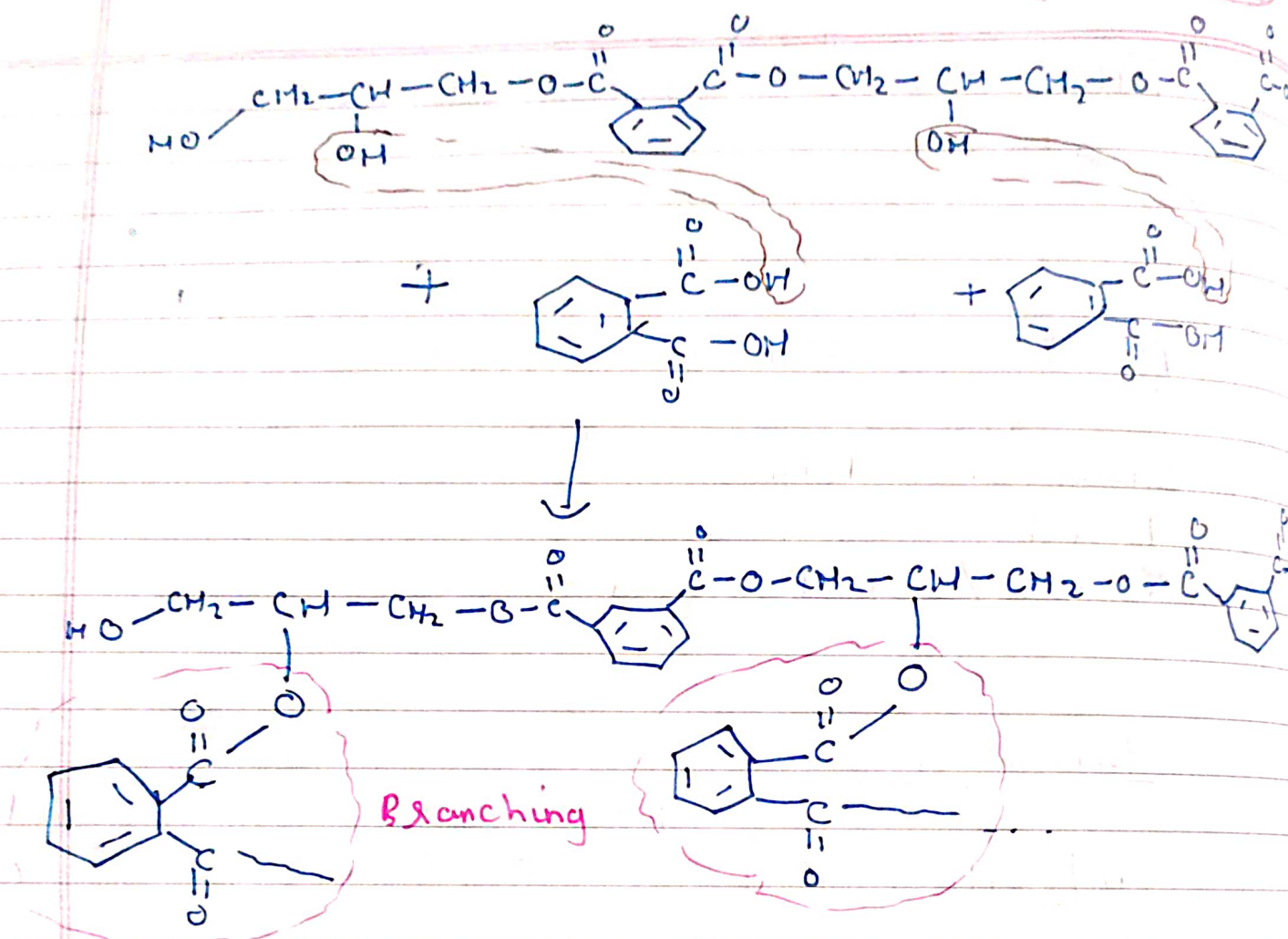


Branch Chain

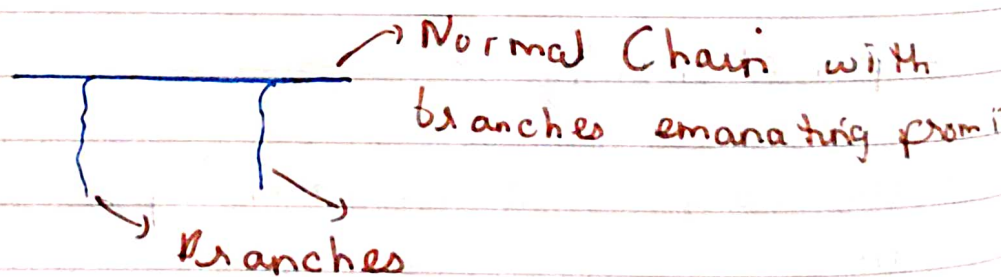
- functionality > 2

- We carry out the rxn for shorter period of time.



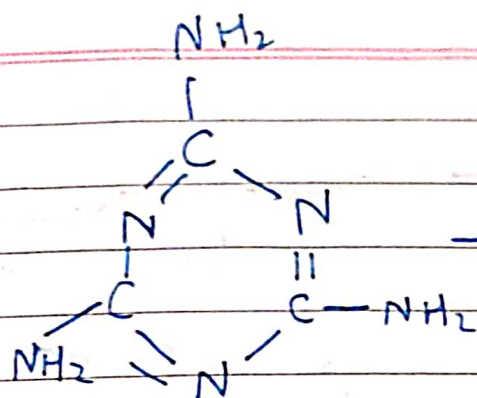


Branched Chain Polymer



* POLYMER FUNDAMENTALS (Cont'd)

* THERMOSETS



white powder
Melamine

Thermal
polymerization
550°C 2h
air atmosphere

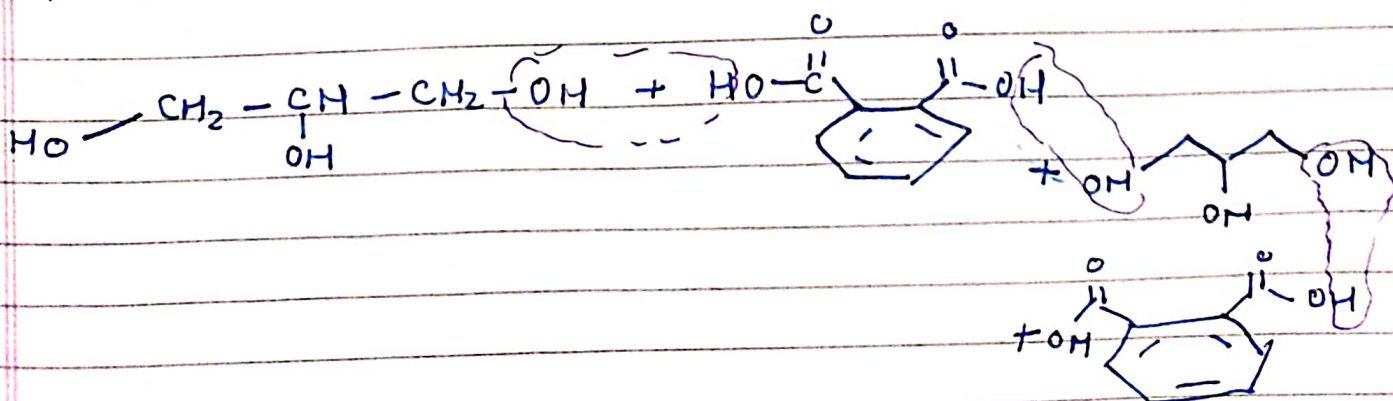
Bulk g-C₃N₄
graphite carbon
nitride
Cyan in colour
metal free
polymer molecule

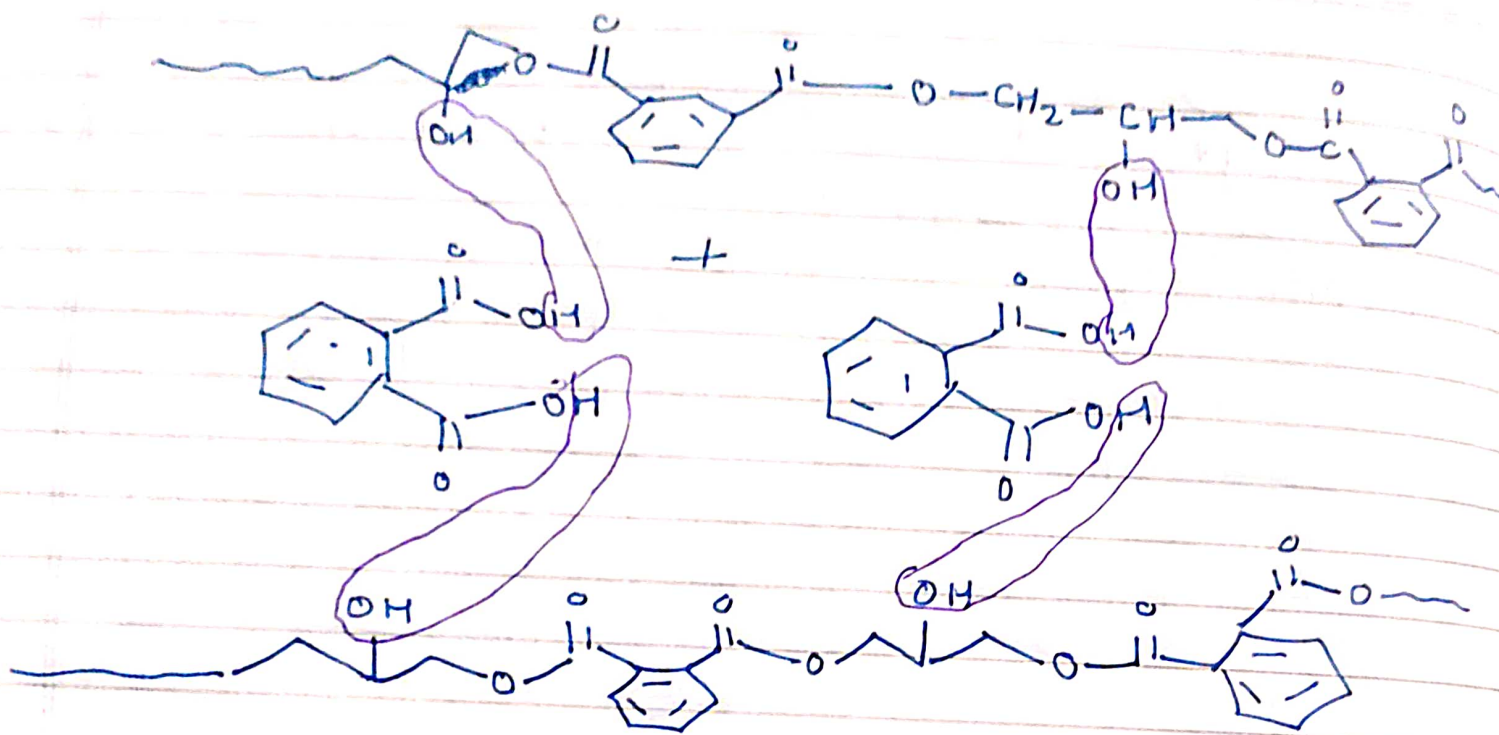
graphitic carbon nitride
g-C₃N₄ (pale yellow)
separate from C₂H₅OH
C₂H₅OH (99.99%)
(i) stirring 1.5h (magnetic)
(ii) ultra sonication 1.5h
(iii) vacuum filtration/centrifugation
(iv) Oven drying

* Network / Gel Formation

(i) Functionality of monomer > 2

ii) We will carry out the reaction for a longer time.





II

Linear chain

Branching

Linear chain

Gel / Network

* POLYETHYLENE PRODUCTION $\text{[-CH}_2\text{-CH}_2\text{]}_n$

→ Properties of Polyethylene

Mol. wt = 1500 to 100,000

Melting point = 85 - 110°C

Density = $\left[\begin{array}{c} 0.91 - 0.93 \\ 0.96 \end{array} \right]$ $\rightarrow$ LDPE $\rightarrow$ Low density Polyethylene
 $\rightarrow$ HDPE $\rightarrow$ High density Polyethylene

→ * THREE DIFF PROCESSES

① High Pressure Process (VCM)

$P = 1000 - 2500 \text{ atm}$

$T = 100 - 300^\circ\text{C}$

Catalyst = peroxide based catalyst $\rightarrow$ LDPE

② Intermediate Pressure Process (CRA)

$P = 30 - 100 \text{ atm}$

$T = < 150^\circ\text{C}$

Catalyst = $\text{Cr}_2\text{O}_3 / \text{Al}_2\text{O}_3$ $\rightarrow$ / SiO₂ / ZrO₂ (Zirconia)
 $\text{MoO}_3 / \text{Al}_2\text{O}_3 \rightarrow$ HDPE

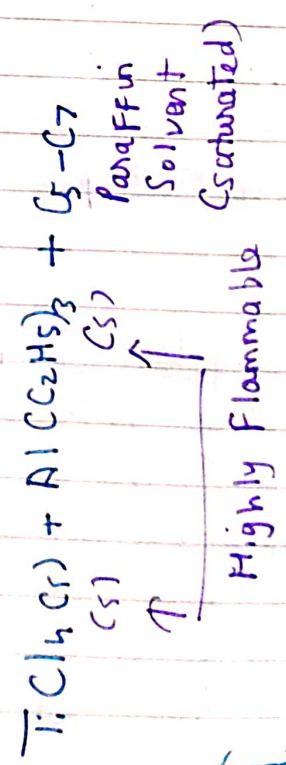
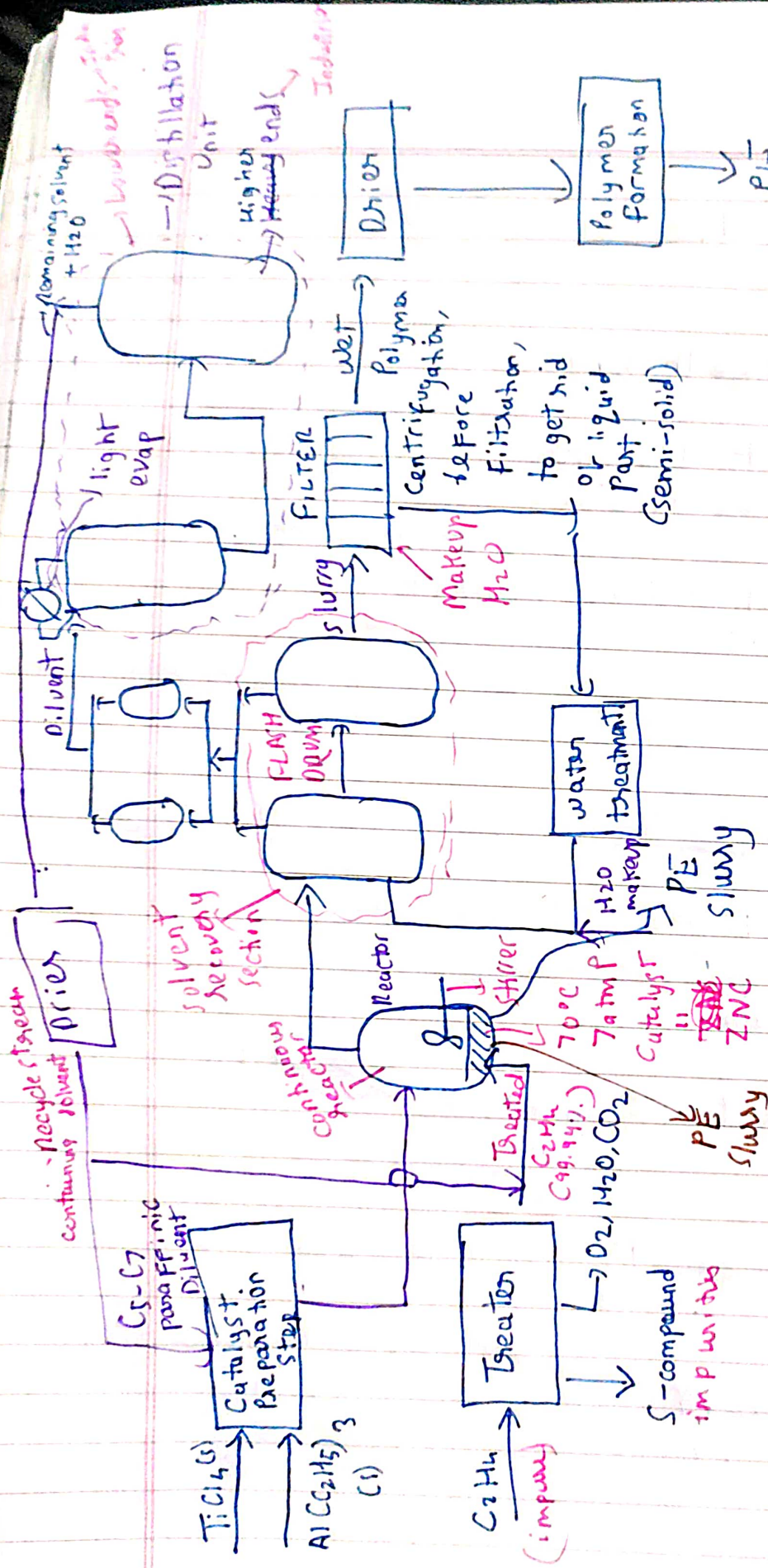
③ Low Pressure Process

$P = 6 - 10 \text{ atm}$

$T = 70^\circ\text{C}$
 C_2H_5
 C_2H_5
 C_2H_5

Catalyst = $\text{TiCl}_4 + \text{Al}$

$\rightarrow$ LDPE
 $+ \text{HDPE}$



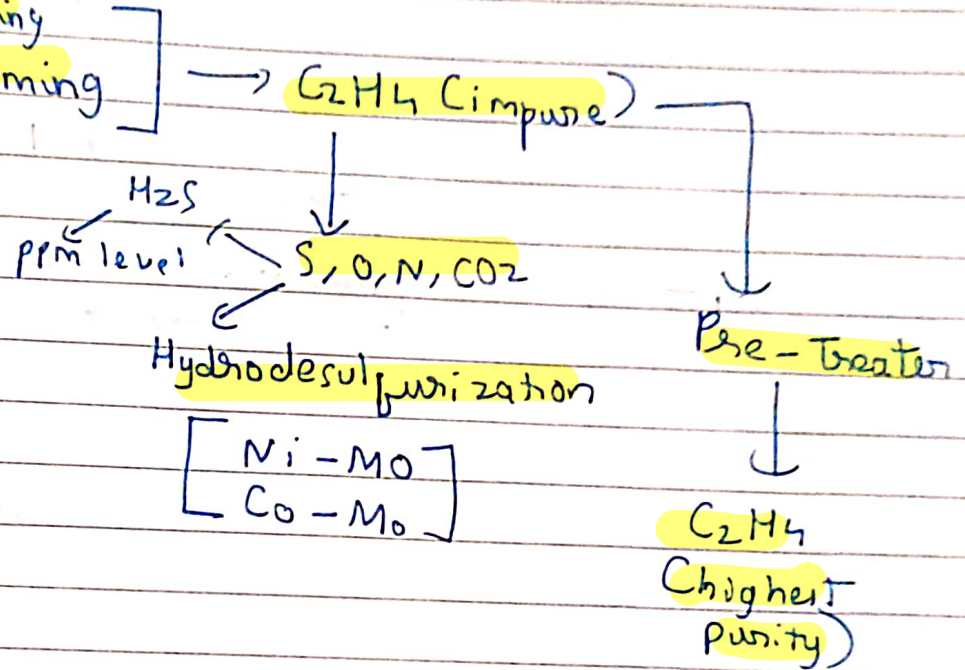
$Cr_2O_3 - MoO_3 / S.O_2$
 $Cr_2O_3 - MoO_3 / S.O_2 - Al_2O_3$

* Sources of C_2H_4

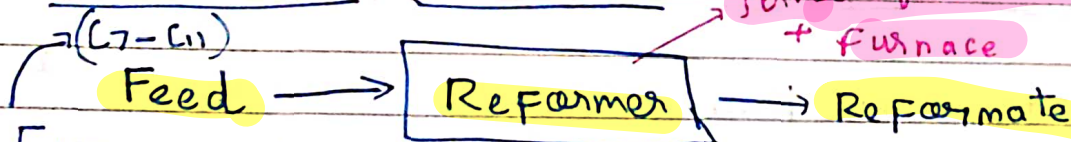
- Petroleum Refinery

a) Catalytic Cracking

b) Catalytic Reforming



* PETROLEUM REFINERY



Fuel
Quality
is not
that
good

↓
n-paraffin compounds
i-paraffin "
olefins
Naphthalene
Aromatics
Hydrogen

octane no: 20-50

(IV) MoO_3/SiO_2
 $MoO_3/SiO_2-Al_2O_3$

(II) Ni/SiO_2
 $Ni/SiO_2-Al_2O_3$

(III) Cr_2O_3/SiO_2
 $Cr_2O_3/SiO_2-Al_2O_3$

$T = 450^\circ C - 500^\circ C$

$P = 10-50 \text{ atm}$

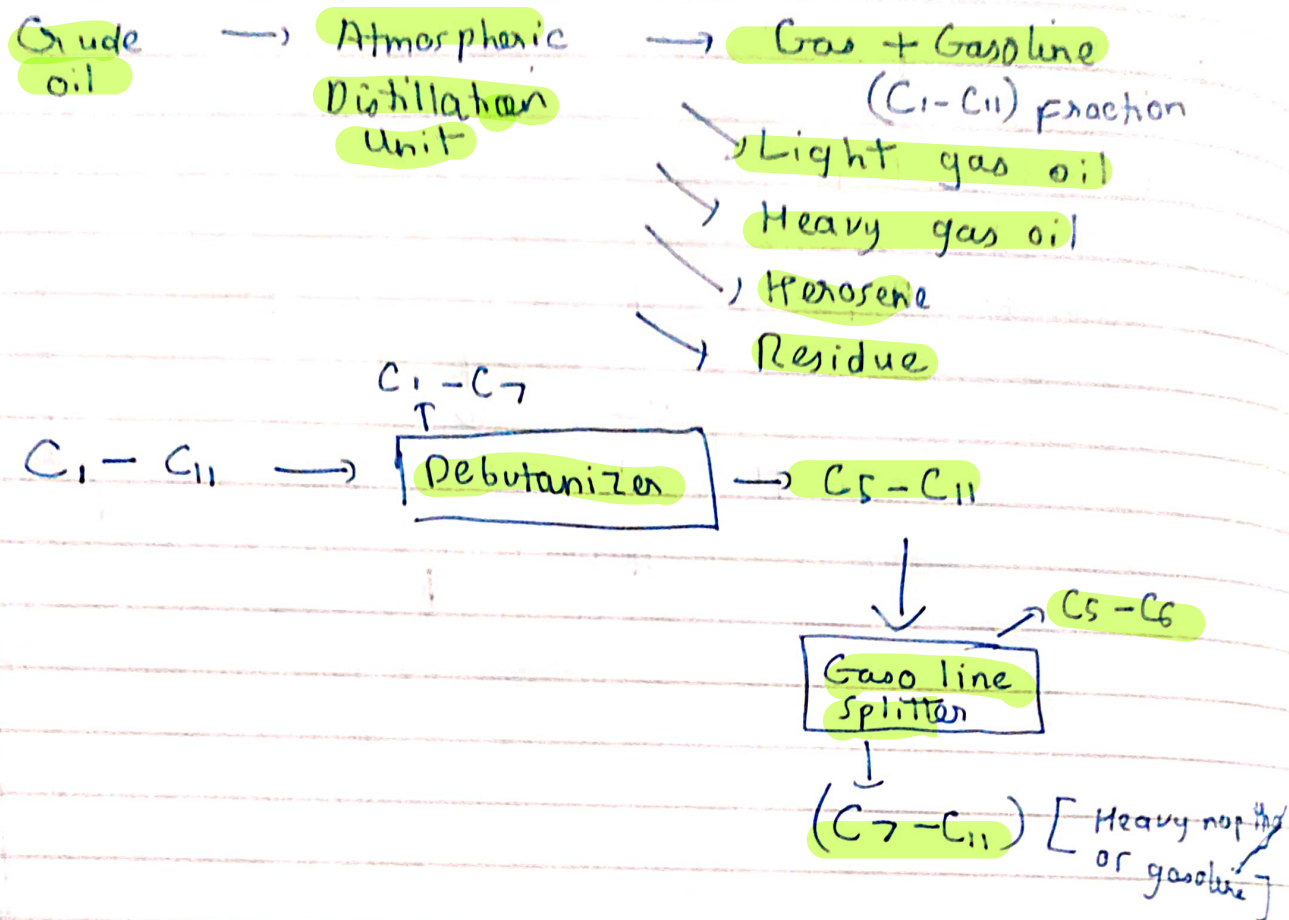
Catalyst: Pt based
Catalyst

(I) Pt/SiO_2
 Pt/SiO_2
Aromatics/Fuel
Quality
is
better

n-paraffin
i-paraffin
olefin
naphthalene
Aromatic
Hydrogen

octane no: 100

More aromatic
compounds



Reactions Performed in the Reformer Unit

- (i) Dehydrogenation Rxn
- ii) Isomerization Rxn
- iii) Paraffin Cracking rxn
- (iv) Hydrogenation rxn
- v) Paraffin Dehydro Cyclization $\rightarrow$ slowest reaction (severe experimental condition)
- vi) Napht^{ene}~~thane~~ isomerization and dehydrogenation
- vii) Hydro desulphurization rxn
- viii) Deoxydation rxn

Exothermic
in nature
< 300°C
Raney Ni

(460-490°C)

Major
rxns

All
rxns
except
hydrogenation
are
ENDO
THERMIC

ix) Denitrogenation rxn

x) Combination rxn

Minor

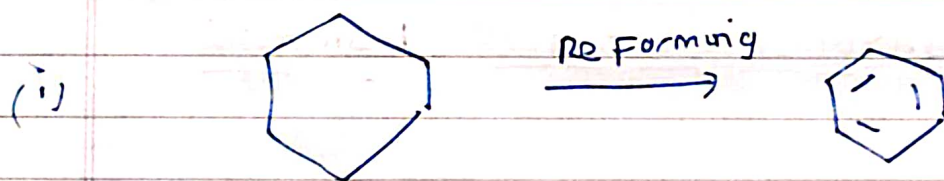
rxns $\rightarrow$ Removal of impurities
(C-S, -O, -N)

Note: In catalyst Pt/SiO_2 or $Pt/SiO_2-Al_2O_3$, if the loading of Pt $\rightarrow$ 0.1 to 0.5% $\rightarrow$ non-regenerative catalyst

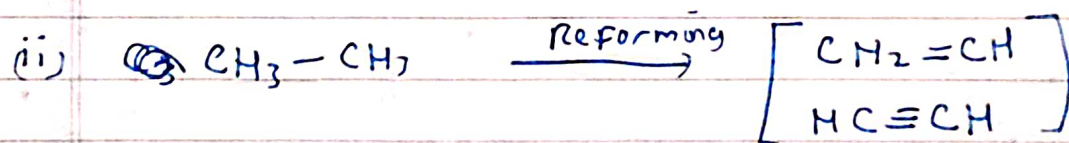
and if loading $\rightarrow$ >0.5% <1% $\rightarrow$ regenerative catalyst

- Platforming $\rightarrow$ Pt as an active catalyst.

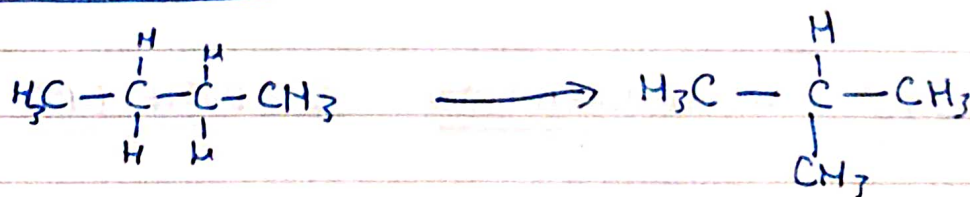
* DEHYDROGENATION REACTION



- For exam, write name of reactant, product, ~~name~~ T, P, catalyst



* ISOMERIZATION REACTION



n-butane

isobutane



n-pentane



neopentane

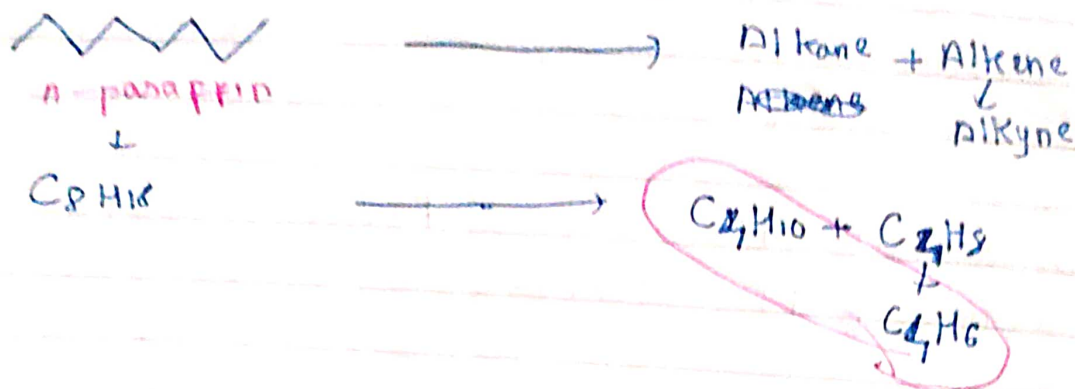
$\rightarrow$ (2,2 dimethyl propane)



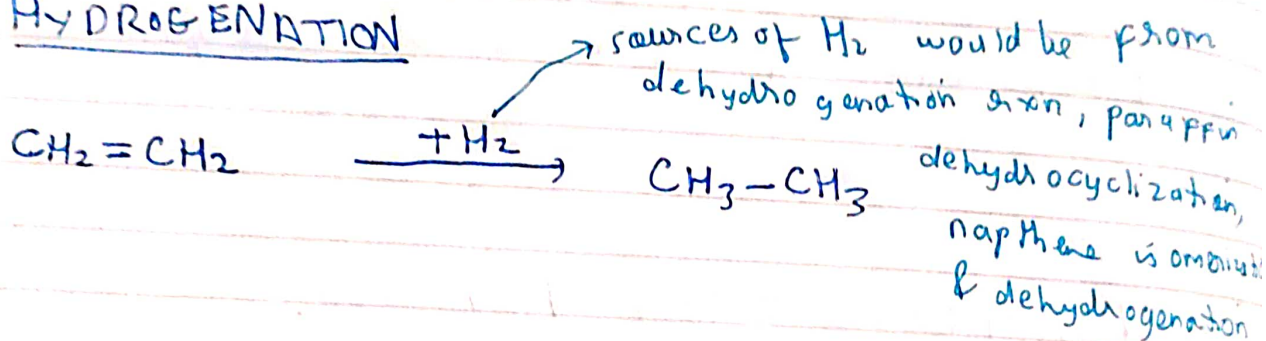
isopentane

$\rightarrow$ (2 methyl butane)

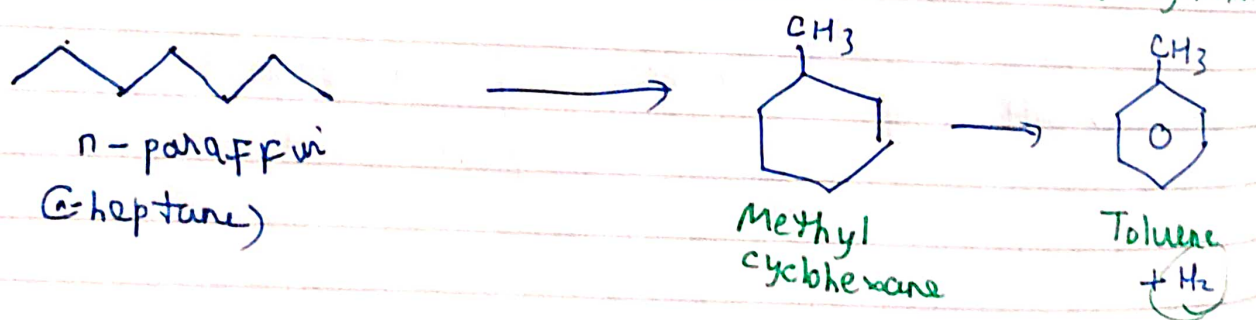
* PARAFFIN CRACKING REACTION



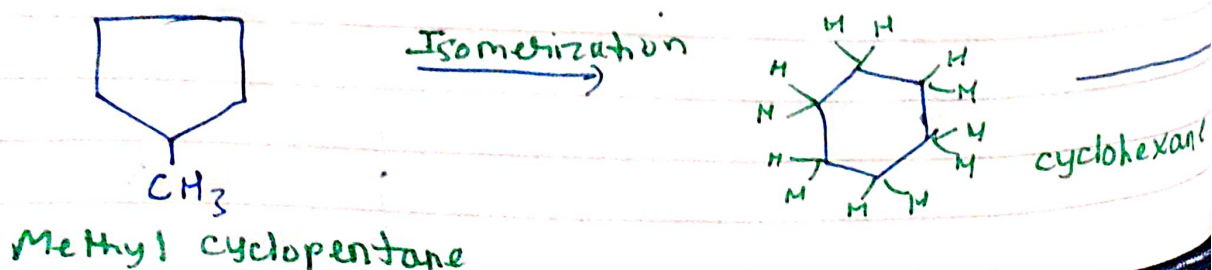
* HYDROGENATION

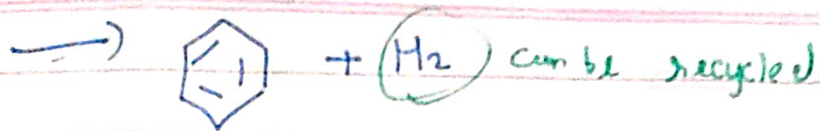


* PARAFFIN DEHYDROCYCLIZATION REACTION

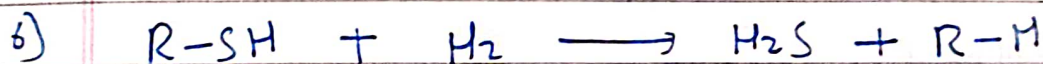
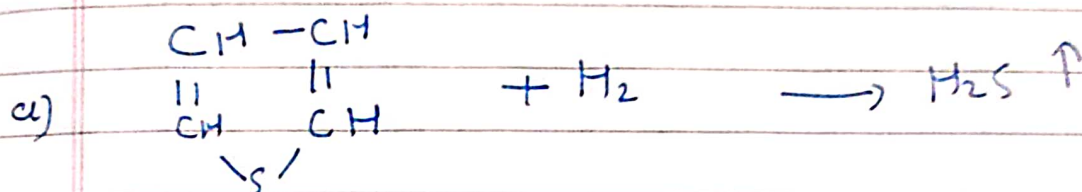


* NAPHTHENE ISOMERIZATION & DEHYDROGENATION

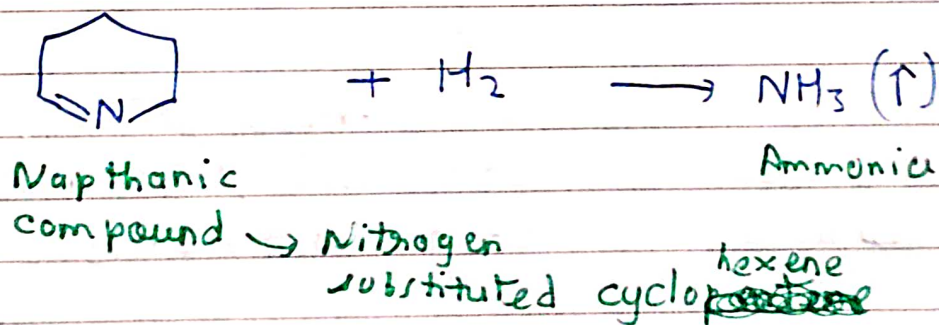




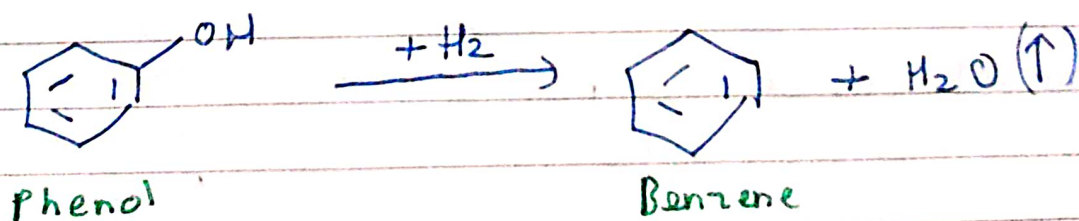
* HYDRODESULFURIZATION REACTION



* DENITROGENATION RXN



* DEOXYDATION RXN



* COMBINATION RXN



Catalyst

Pt / SiO₂
 Pt / SiO₂ - Al₂O₃
 Pt / Al₂O₃

Ni / SiO ₂	Cr ₂ O ₃ / SiO ₂
Ni / SiO ₂ - Al ₂ O ₃	Cr ₂ O ₃ / SiO ₂ - Al ₂ O ₃
Ni / Al ₂ O ₃	Cr ₂ O ₃ / Al ₂ O ₃

MoO₃ / SiO₂
 MoO₃ / SiO₂ - Al₂O₃
 MoO₃ / ~~SiO₂~~ Al₂O₃

Cr₂O₃ - MoO₃ / SiO₂
 Cr₂O₃ - MoO₃ / SiO₂ - Al₂O₃
 Cr₂O₃ - MoO₃ / ~~SiO₂~~ Al₂O₃

90% reactions → fixed bed

10% reactions → fluidised bed / moving bed

Catalyst should have features:

① Acidic surface sites

② Metallic bonds

↓
Facilitate the hydrogenation of

↓
facilitate the cyclization and isomerization of

↓
TPD probe with NH₃

used for regeneration of catalyst. It has moving parts which can be taken out

- If the acidic sites are not sufficient for the platfoming ~~rxn~~^{operation}, we carry out halogenation rxn (C-Cl)

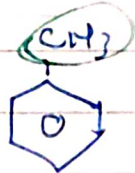
↳ 0.1 wt% Cl loading - → optimised loading of Cl

- If we increase/decrease the pressure and go beyond the 10-50 atm range then:-

- $P < 10 \text{ atm}$ → Coking can take place

↓
Catalyst surrounded by C particles

- $P > 50 \text{ atm}$

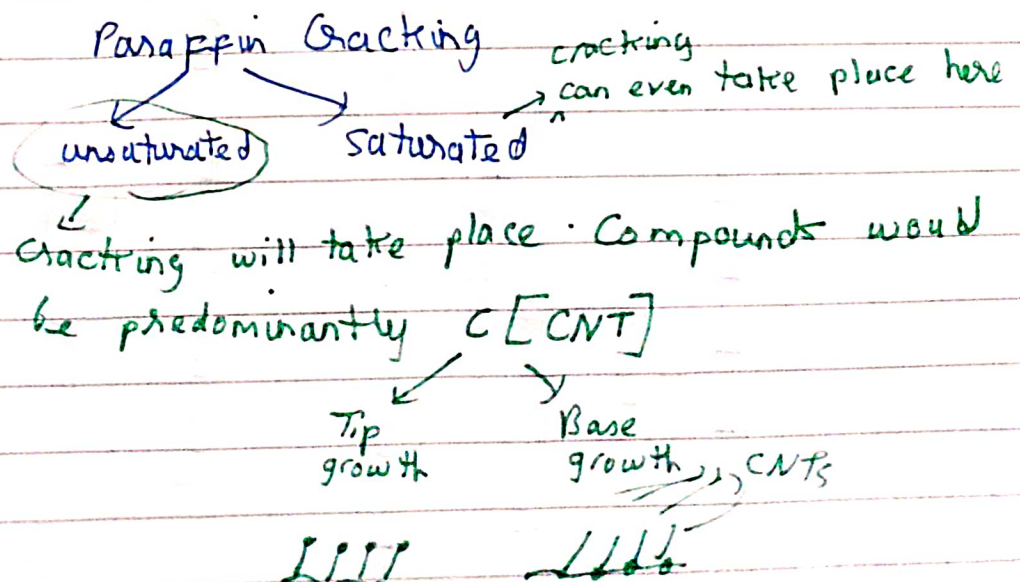


→ dealkylation rxn will take place

→ Effect of T:- Apart from Hydrogenation R^n , all the other 9 are endothermic. Hence, overall process is endothermic.

- At low Temp:- no rxn

- At high Temp:- Coking
>500°C



Catalyst $\rightarrow$ pellet form
(0.3 mm diameter)

we can achieve 97-98% conv by this process.

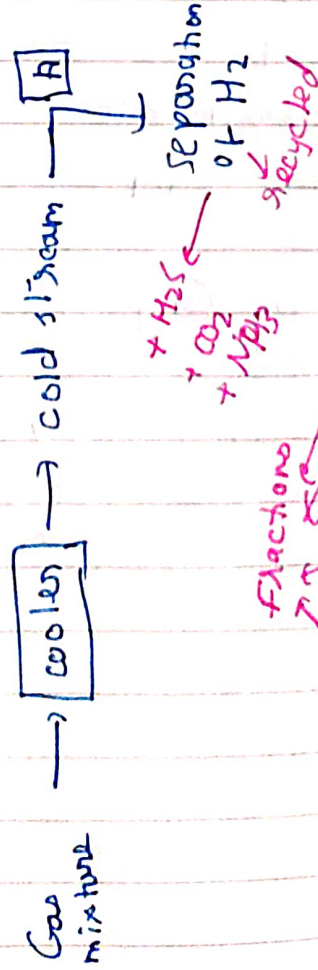
F_1, F_2, F_3, F_4

$\downarrow$
Furnace

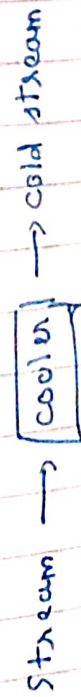
A $\Rightarrow$ high pressure separator
 $\nearrow$ changed stock
 $\nearrow$ heavy naphtha / gasoline

Feed Composition (C/S + Recycle H₂)

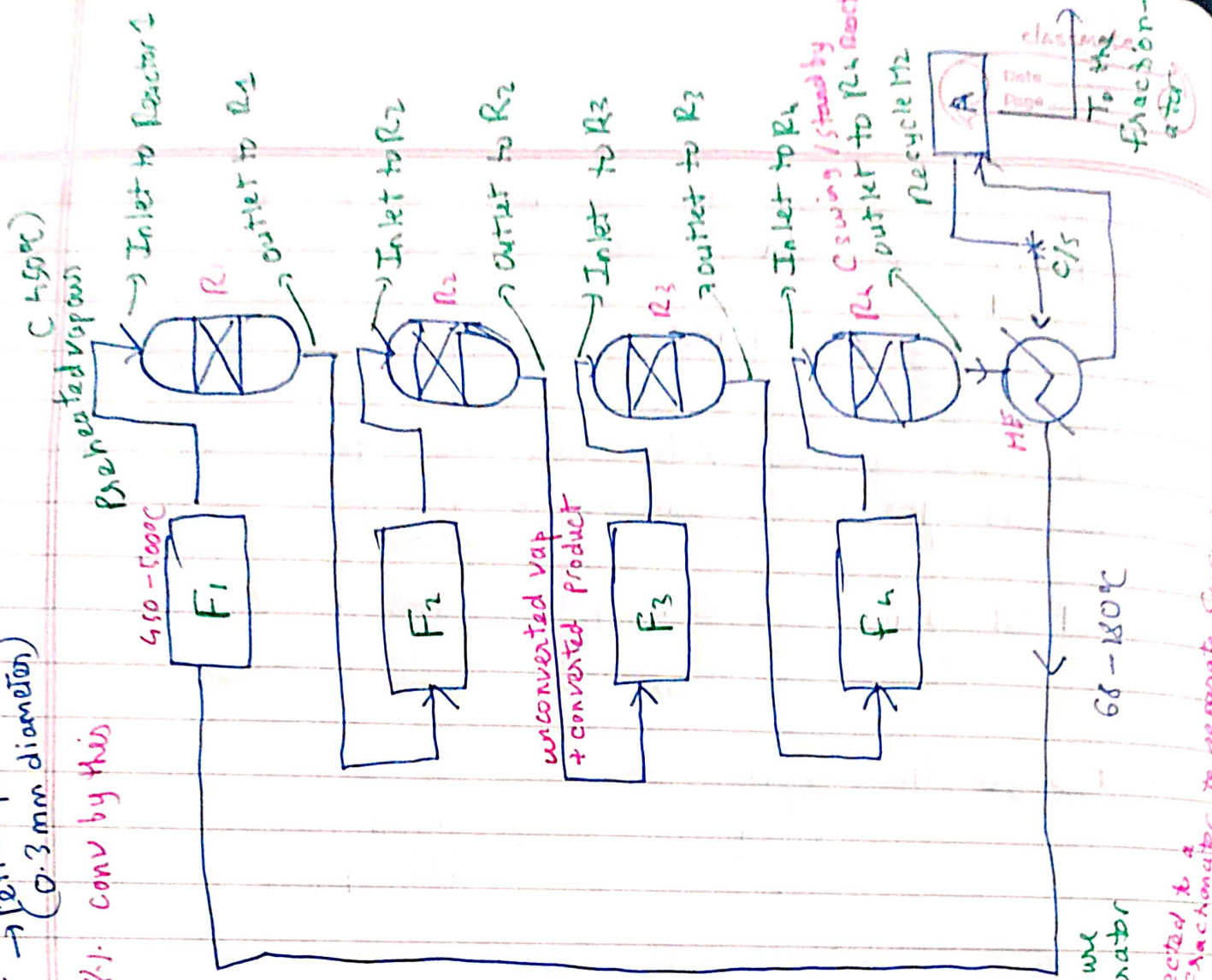
* Separation of H₂ from a gaseous mixture



* Separation of C₁, C₂, C₃, C₄...



$\rightarrow$ low pressure separator
 $\downarrow$ connected to a fractionator to separate C₁, C₂, C₃



— We mainly use R_1 , R_2 and R_3 . R_4 is mostly used as a stand by reactor.

— $R_1 \rightarrow$ low Δx ns

$R_2 \rightarrow$ Isomerization, Paraffin cracking..

$R_3 \rightarrow$ Cyclization

$\hookrightarrow$ slowest Δn which requires extreme condition (460-490°C)

reactionator to separate C₁, C₂, C₃...